```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv(r"C:\Users\nitin\Desktop\python\project\EDA Project\
SuperStore_Management_System.csv")

df.head(20)
```

```
    order Id  order Date   Ship Date Customer ID      Customer Name  \
0   ORD10000  2024-05-21  2024-05-28     CUST730      Brooke Wright
1   ORD10001  2025-11-19  2025-11-22     CUST659         Mary Mills
2   ORD10002  2025-02-23  2025-03-02     CUST250        Jason Weiss
3   ORD10003  2025-03-21  2025-03-23     CUST711       Sharon Smith
4   ORD10004  2026-03-29  2026-04-02     CUST612        John Henson
5   ORD10005  2024-10-14  2024-10-21     CUST982     Stephen Newton
6   ORD10006  2024-11-18  2024-11-21     CUST564       Lisa Everett
7   ORD10007  2024-10-28  2024-10-30     CUST958        Jacob Dixon
8   ORD10008  2025-03-31  2025-04-07     CUST916        Tammy Moore
9   ORD10009  2024-08-01  2024-08-05     CUST667        Amy Hensley
10  ORD10010  2025-10-11  2025-10-12     CUST150       Eric Roberts
11  ORD10011  2024-06-10  2024-06-12     CUST945  Barbara Pratt DVM
12  ORD10012  2025-12-19  2025-12-21     CUST681         James Cole
13  ORD10013  2025-09-01  2025-09-05     CUST502   Christine Haynes
14  ORD10014  2025-03-04  2025-03-05     CUST529    Melanie Shaffer
15  ORD10015  2024-11-29  2024-12-03     CUST853       Amber Brooks
16  ORD10016  2025-02-12  2025-02-18     CUST579      John Anderson
17  ORD10017  2025-10-24  2025-10-30     CUST497       Gregory Moss
18  ORD10018  2025-12-06  2025-12-07     CUST337      Jessica Adams
19  ORD10019  2025-12-23  2025-12-26     CUST792       Mariah Casey

   Customer Segment Product ID  Product Name         Category
Region  ...  \
0          Consumer  PROD18948    Calculater  Office Supplies
South  ...
1         Corporate  PROD17245      BookSelf        Furniture
West  ...
2       Home Office  PROD15496      Keyboard      Electronics
North  ...
3          Consumer  PROD17331      BookSelf        Furniture
North  ...
4       Home Office  PROD13838  Office Chair        Furniture
North  ...
5         Corporate  PROD19720    Headphones      Electronics
West  ...
6         Corporate  PROD12042      BookSelf        Furniture
South  ...
7       Home Office  PROD13374        Snacks          Grocery
North  ...
8          Consumer  PROD11235      Notebook  Office Supplies
```

```
North  ...
9         Corporate  PROD16521   Cooking Oil          Grocery
South  ...
10        Corporate  PROD15549   File Folder  Office Supplies
North  ...
11         Consumer  PROD19173          Sofa        Furniture
South  ...
12      Home Office  PROD13485           Pen  Office Supplies
North  ...
13        Corporate  PROD11515        Snacks          Grocery
North  ...
14      Home Office  PROD18097      Rice Bag          Grocery
North  ...
15        Corporate  PROD14747  Dining Table        Furniture
West  ...
16        Corporate  PROD18860   Cooking Oil          Grocery
North  ...
17         Consumer  PROD19479   File Folder  Office Supplies
South  ...
18         Consumer  PROD16649   Study Table        Furniture
North  ...
19        Corporate  PROD19168       Stapler  Office Supplies
West  ...

   Sales Amount Cost Price    Profit  Payment Mode  Delivery Status  \
0       1165.35     810.49    354.86          Cash         Returned
1      62402.40   34143.94  28258.46   Credit Card        Delivered
2     144171.00  129094.94  15076.06           UPI         Returned
3      14229.00    9396.89   4832.11   Credit Card         Returned
4      40882.50   31687.33   9195.17           UPI        Delivered
5      79449.60   72828.29   6621.31   Net Banking        Cancelled
6      12500.95    8630.34   3870.61   Credit Card         Returned
7        691.00     578.02    112.98   Net Banking         Returned
8       2822.40    1590.85   1231.55          Cash        Delivered
9       2802.50    2477.97    324.53          Cash        Cancelled
10       370.80     214.61    156.19           UPI        Cancelled
11     29556.00   16870.45  12685.55           UPI          Pending
12      1581.00     774.73    806.27          Cash        Delivered
13      1074.00     849.41    224.59          Cash        Delivered
14      1158.30     925.96    232.34   Credit Card          Pending
15     91106.40   69567.42  21538.98           UPI        Cancelled
16      1547.00    1369.98    177.02           UPI          Pending
17      5502.40    2530.06   2972.34           UPI          Pending
18     23319.00   11704.74  11614.26          Cash        Delivered
19     10696.40    7162.81   3533.59           UPI        Cancelled

                      Supplier Name                      Supplier Email
\
0                       Wong-Pierce    bernarddaniel@chang-murillo.com
```

1	Jackson-Hogan	burnettashley@daniel.biz
2	Carrillo-Thomas	jwall@calhoun-davidson.com
3	Bates, Smith and Turner	xstokes@parker.com
4	Cooper, Davies and Pugh	lindamccoy@morton.com
5	Williams-Hill	marissafitzpatrick@green-ray.com
6	Reynolds-Perez	fharris@barrett.biz
7	Schmidt, Mora and Simmons	theresa73@peterson.net
8	Montgomery, Moore and Combs	crichards@harvey-vazquez.com
9	Cruz and Sons	mbrown@edwards.com
10	Harris Inc	chavezmarie@snyder.info
11	Mueller-Webb	kelseymartin@benjamin.com
12	Lin, Estrada and Torres	bruce44@edwards.com
13	Brown, Manning and Ashley	carmen81@miller-mason.com
14	Mccoy-Miller	tyler74@gaines.org
15	Gallagher, Hammond and Hensley	lydia93@webster.com
16	Lloyd, Mills and Herman	mturner@soto.com
17	Henry-Fletcher	caitlincross@williams-aguilar.com
18	Pearson LLC	melissa01@cole.biz
19	Jones PLC	erin29@neal.info

	Stock Left	Auto Reorder	Reorder Quantity
0	49	No	0
1	33	No	0
2	38	No	0
3	38	No	0
4	41	No	0
5	46	No	0
6	46	No	0
7	36	No	0
8	21	No	0
9	25	No	0
10	49	No	0

```
11          23           No                 0
12          27           No                 0
13          18           No                 0
14           7          Yes                33
15           6          Yes                44
16          47           No                 0
17          37           No                 0
18          18           No                 0
19          13           No                 0

[20 rows x 25 columns]
```

```
total_Sales = df["Sales Amount"].sum()
total_profit = df["Profit"].sum()

print("total Sales :",total_Sales)
print("total profit :" ,total_profit)
```

```
total Sales : 563351500.4000001
total profit : 102206295.34
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 25 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   order Id          10000 non-null  str
 1   order Date        10000 non-null  str
 2   Ship Date         10000 non-null  str
 3   Customer ID       10000 non-null  str
 4   Customer Name     10000 non-null  str
 5   Customer Segment  10000 non-null  str
 6   Product ID        10000 non-null  str
 7   Product Name      10000 non-null  str
 8   Category          10000 non-null  str
 9   Region            10000 non-null  str
 10  State             10000 non-null  str
 11  City              10000 non-null  str
 12  Quantity          10000 non-null  int64
 13  Unit Price        10000 non-null  int64
 14  Discount (%)      10000 non-null  int64
 15  Sales Amount      10000 non-null  float64
 16  Cost Price        10000 non-null  float64
 17  Profit            10000 non-null  float64
 18  Payment Mode      10000 non-null  str
 19  Delivery Status   10000 non-null  str
 20  Supplier Name     10000 non-null  str
 21  Supplier Email    10000 non-null  str
```

```
 22  Stock Left        10000 non-null  int64
 23  Auto Reorder      10000 non-null  str
 24  Reorder Quantity  10000 non-null  int64
dtypes: float64(3), int64(5), str(17)
memory usage: 1.9 MB
```

```
df["Sales Amount"] = df["Sales Amount"].astype(int)
```

```
df["Profit"] = df["Profit"].astype(int)
df["Cost Price"] = df["Cost Price"].astype(int)
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 25 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   order Id          10000 non-null  str
 1   order Date        10000 non-null  str
 2   Ship Date         10000 non-null  str
 3   Customer ID       10000 non-null  str
 4   Customer Name     10000 non-null  str
 5   Customer Segment  10000 non-null  str
 6   Product ID        10000 non-null  str
 7   Product Name      10000 non-null  str
 8   Category          10000 non-null  str
 9   Region            10000 non-null  str
 10  State             10000 non-null  str
 11  City              10000 non-null  str
 12  Quantity          10000 non-null  int64
 13  Unit Price        10000 non-null  int64
 14  Discount (%)      10000 non-null  int64
 15  Sales Amount      10000 non-null  int64
 16  Cost Price        10000 non-null  int64
 17  Profit            10000 non-null  int64
 18  Payment Mode      10000 non-null  str
 19  Delivery Status   10000 non-null  str
 20  Supplier Name     10000 non-null  str
 21  Supplier Email    10000 non-null  str
 22  Stock Left        10000 non-null  int64
 23  Auto Reorder      10000 non-null  str
 24  Reorder Quantity  10000 non-null  int64
dtypes: int64(8), str(17)
memory usage: 1.9 MB
```

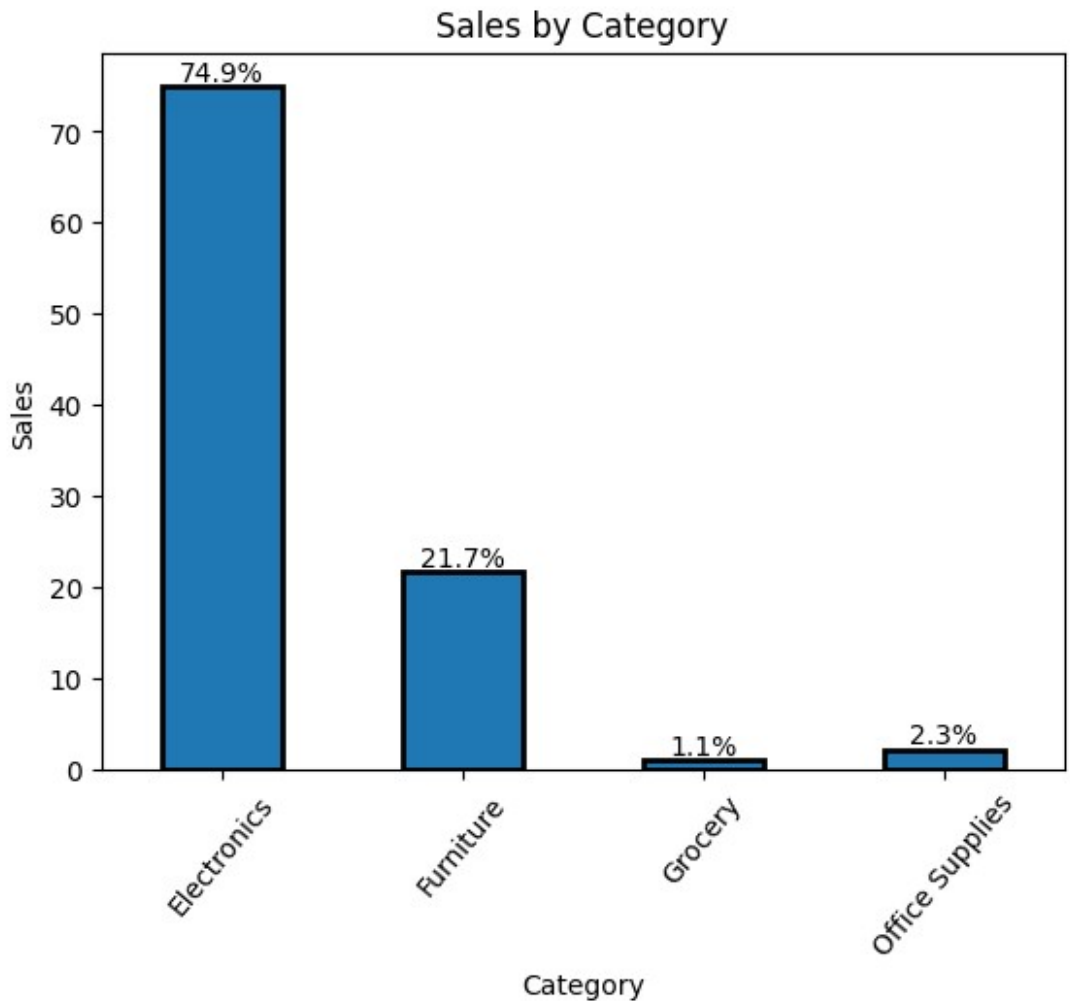
```
df.isnull().sum()
```

```
order Id            0
order Date          0
Ship Date           0
```

```
Customer ID          0
Customer Name        0
Customer Segment     0
Product ID           0
Product Name         0
Category             0
Region               0
State                0
City                 0
Quantity             0
Unit Price           0
Discount (%)         0
Sales Amount         0
Cost Price           0
Profit               0
Payment Mode         0
Delivery Status      0
Supplier Name        0
Supplier Email       0
Stock Left           0
Auto Reorder         0
Reorder Quantity     0
dtype: int64
```

Category wise sales

```
cat_sales = df.groupby("Category")["Sales Amount"].sum()
cat_sales_percentage = (cat_sales / cat_sales.sum())*100
cat_sales_percentage

plt.figure()

ax = cat_sales_percentage.plot(kind = "bar",
                          linewidth = 2,
                          edgecolor = "black")
plt.title("Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")
ax.bar_label(ax.containers[0], fmt = '%1.1f%%')
plt.xticks(rotation = 50)
plt.show()
```



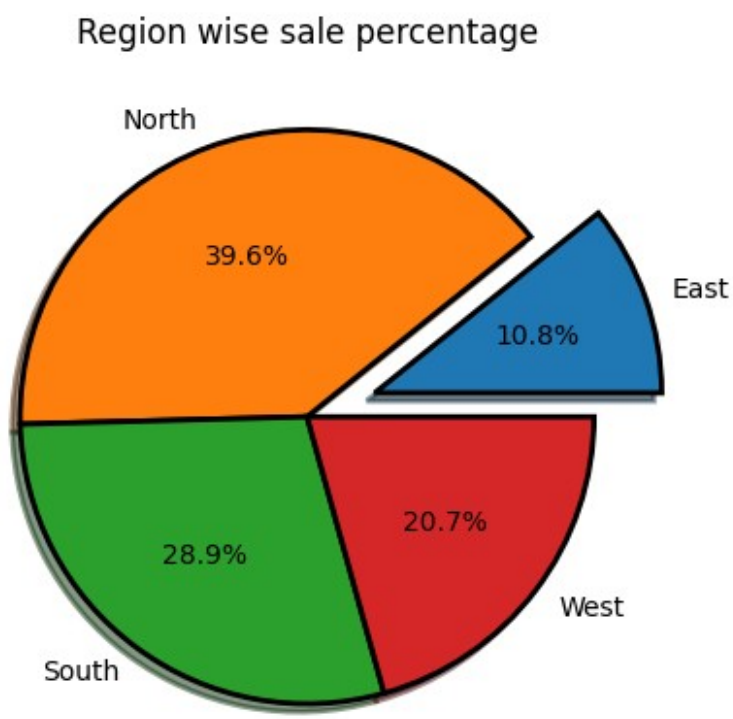
Electronics category contributes the highest share of total sales (~75%) and Grocery and Office Supplies contribute very low sales.

This indicates that the business is highly dependent on Electronics category.

Region wise sale percentage

```
kl = df.groupby("Region")["Sales Amount"].sum()
kl_Percentage = (kl/kl.sum())*100
kl_Percentage

plt.figure()

kl_Percentage.plot(kind = 'pie',
                   labels = kl.index,
                   autopct = "%1.1f%%",
                   explode =[0.25,0,0,0],
                   wedgeprops = {"linewidth" : 2, "edgecolor" :
"black"},
                   shadow = True)
```

```
plt.title("Region wise sale percentage")
plt.show()
```

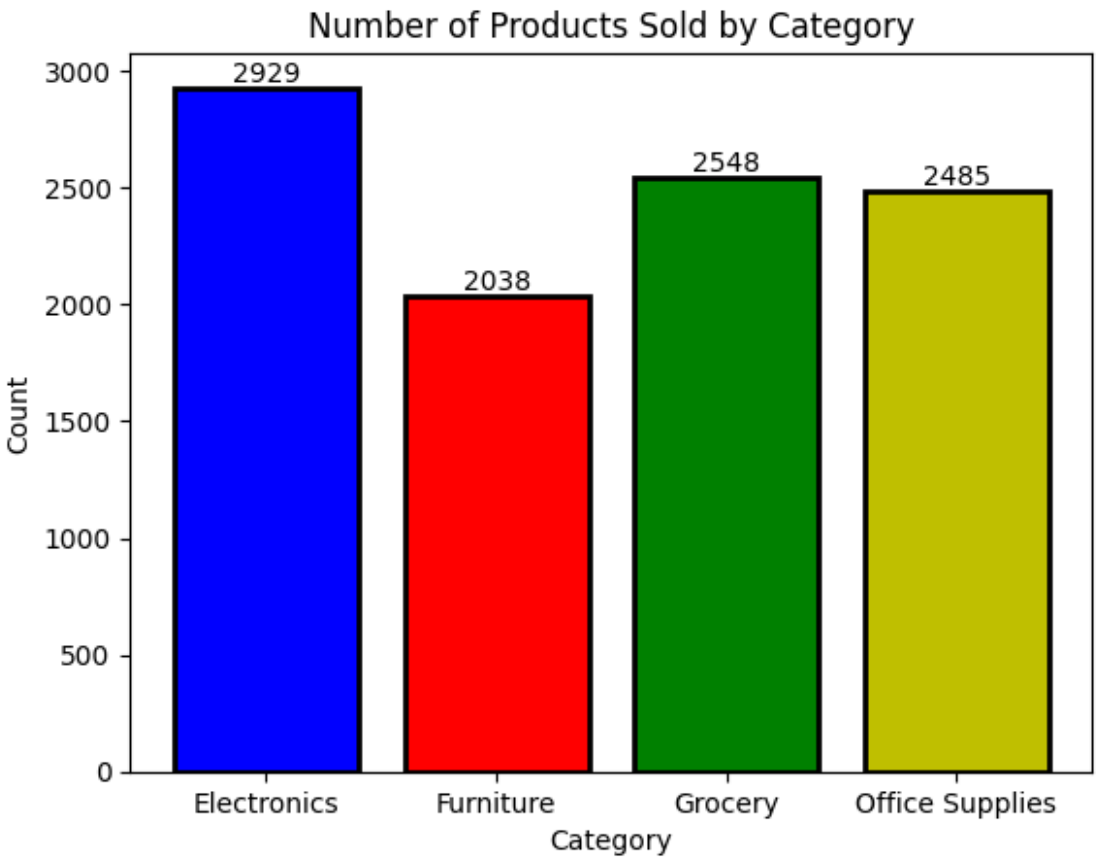


North region is the strongest market for the business.

East region shows low performance

Category wise product sale

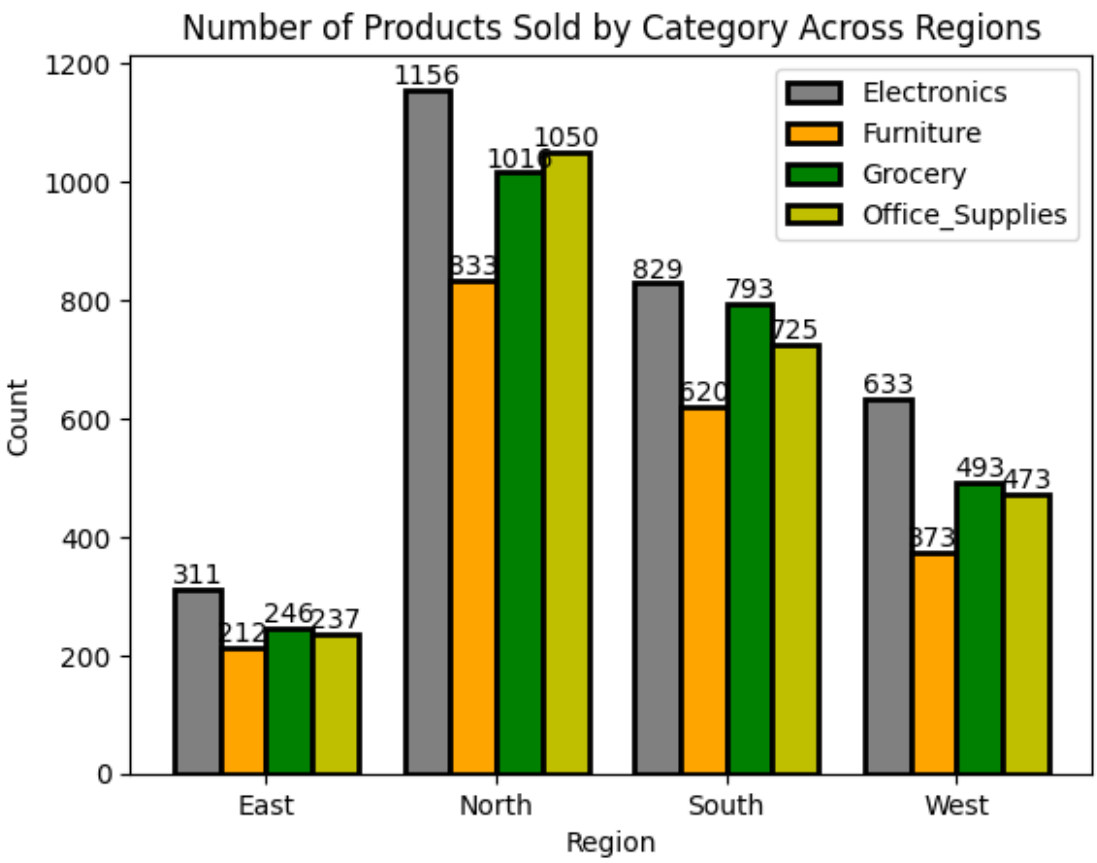
```
kk = df.groupby("Category")["Category"].count()
c = ["b", "r","g", "y"]
x = plt.bar(kk.index, kk.values, edgecolor ="black", linewidth = 2,
color = c)
plt.title("Number of Products Sold by Category")
plt.xlabel("Category")
plt.ylabel("Count")
plt.bar_label(x)
plt.show()
```



Electronics category shows high demand

```
lk = df.groupby(["Region","Category"]).size().unstack()
Reigon = lk.index
Electronics = lk["Electronics"]
Furniture = lk["Furniture"]
Grocery = lk["Grocery"]
Office_Supplies = lk["Office Supplies"]

lk

x = np.arange(len(Category))

width = 0.2

bar1 = plt.bar(x-1.5*width , Electronics, width, edgecolor ="black",
linewidth = 2, color = "gray",label ="Electronics")
bar2 = plt.bar(x-0.5*width , Furniture, width,  edgecolor ="black",
linewidth = 2, color = "orange",label ="Furniture")
bar3 = plt.bar(x+0.5*width , Grocery, width, edgecolor ="black",
linewidth = 2, color = "g",label ="Grocery")
bar4 = plt.bar(x+1.5*width , Office_Supplies, width, edgecolor
="black", linewidth = 2, color = "y",label ="Office_Supplies")
```

```
plt.xticks(x, Reigon)
plt.title("Number of Products Sold by Category Across Regions")
plt.xlabel("Region")
plt.ylabel("Count")

plt.bar_label(bar1)
plt.bar_label(bar2)
plt.bar_label(bar3)
plt.bar_label(bar4)

plt.legend()
plt.show()
```



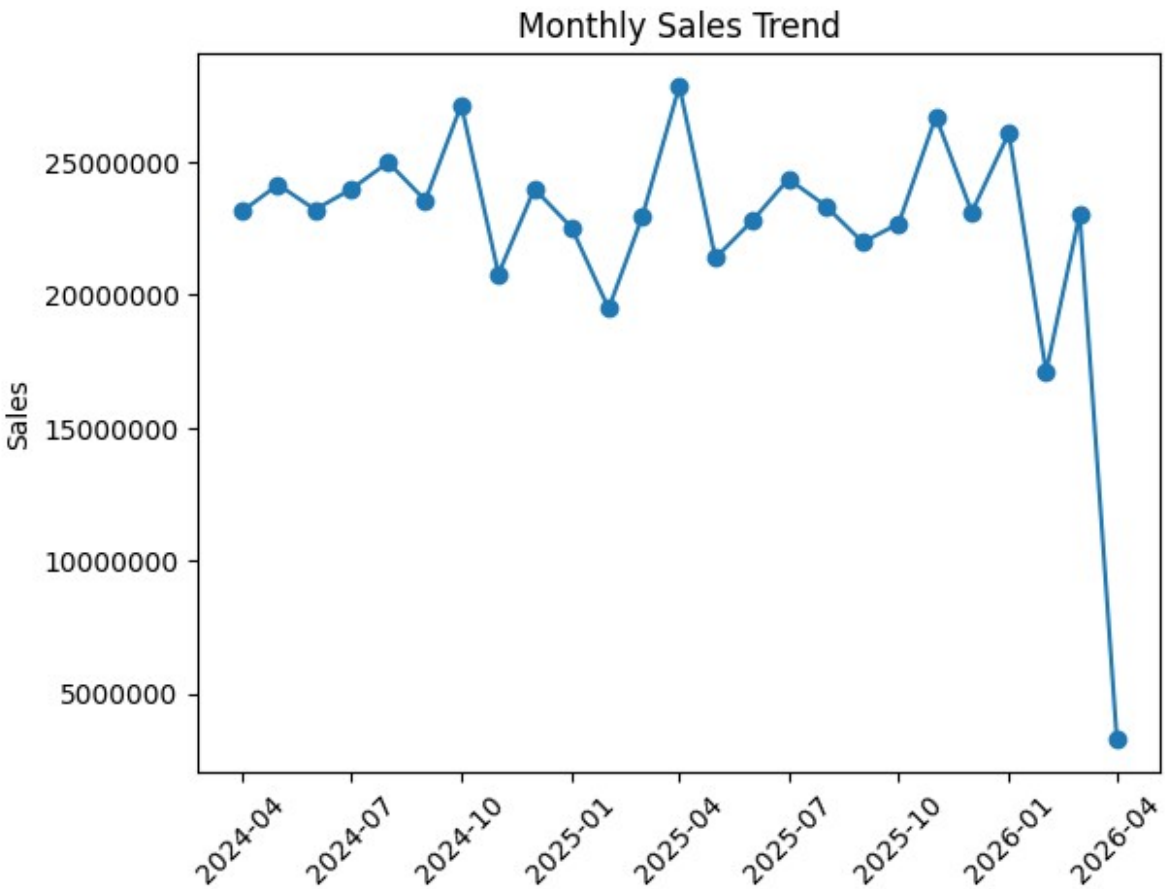
North region has the highest sales in all categories.

East region has the lowest sales and can be improved in the future.

Monthly Trend Analysis

```
df["order Date"] = pd.to_datetime(df["order Date"])
df["Month"] = df["order Date"].dt.to_period("M")

monthly_sales = df.groupby("Month")["Sales Amount"].sum()
```

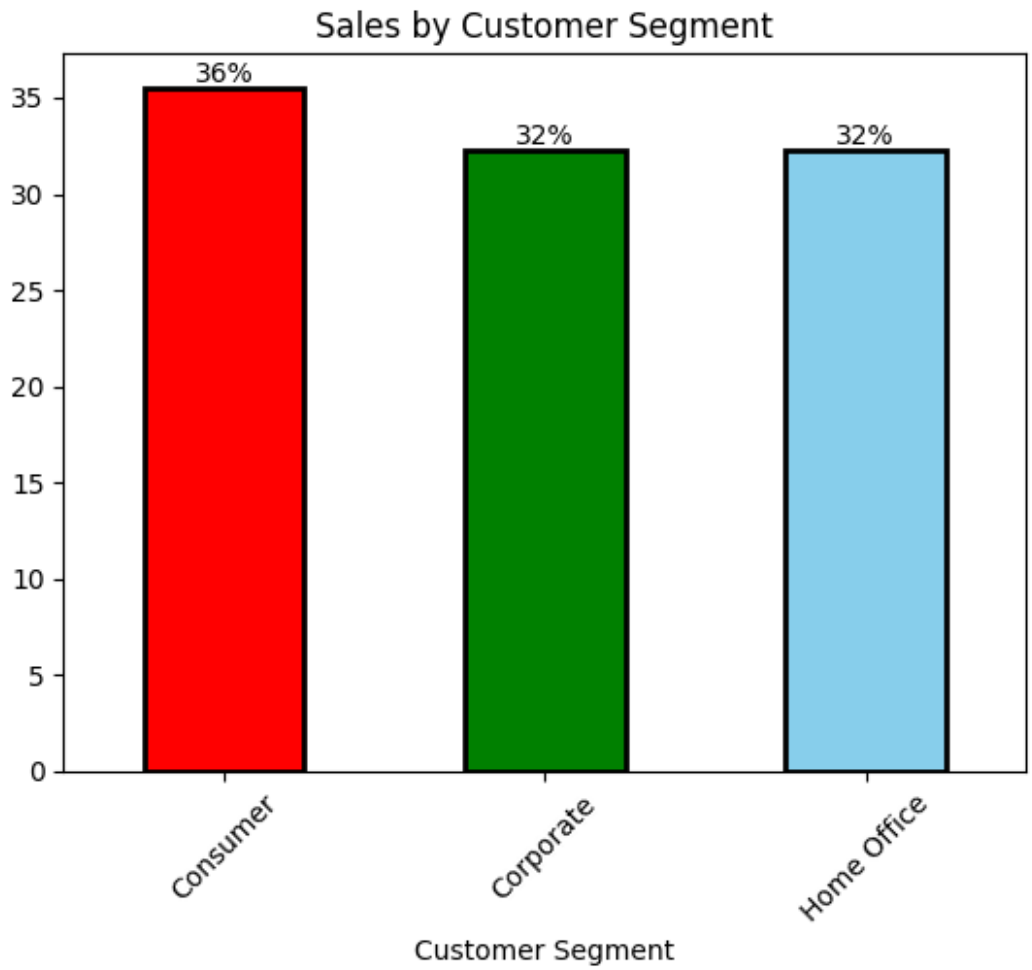
```
# monthly_sales_per = (monthly_sales/monthly_sales.sum())*100
# monthly_sales_per

plt.figure()
plt.plot(monthly_sales.index.to_timestamp(),
monthly_sales.values,marker = "o")
plt.title("Monthly Sales Trend")
plt.xticks(rotation=45)
plt.ylabel("Sales")
plt.ticklabel_format(style='plain', axis='y')
plt.show()
```



Sales show fluctuations over time and Sales drop suddenly at the end.

```
segment_sales = df.groupby("Customer Segment")["Sales Amount"].sum()

segment_sales_per = (segment_sales/segment_sales.sum())*100
segment_sales_per
plt.figure()
x = segment_sales_per.plot(kind = "bar",
                           linewidth = 2,
                           edgecolor = "black",
```

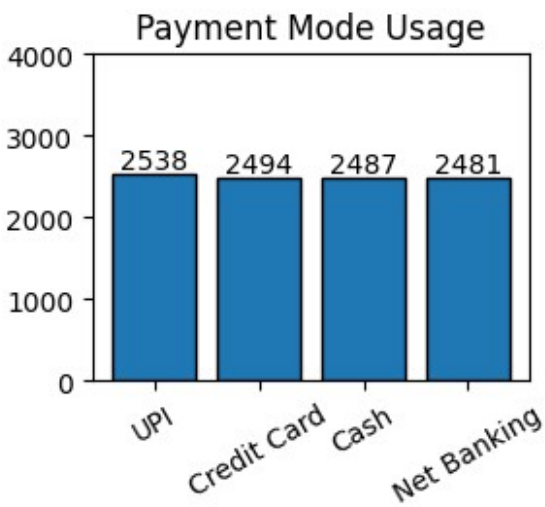
```
                              color = ["r","g","skyblue"])
plt.title("Sales by Customer Segment")
x.bar_label(x.containers[0], fmt = "%1.0f%%")
plt.xticks(rotation = 45)

plt.show()
```

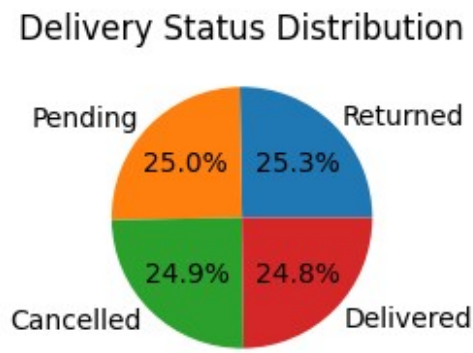


payment mode analysis

```
payment_counts = df["Payment Mode"].value_counts()



plt.subplot(2,2,1)
bar1 = plt.bar(payment_counts.index, payment_counts.values,
edgecolor="black")

plt.title("Payment Mode Usage")
plt.xticks(rotation=30)
plt.ylim(0,4000)
```

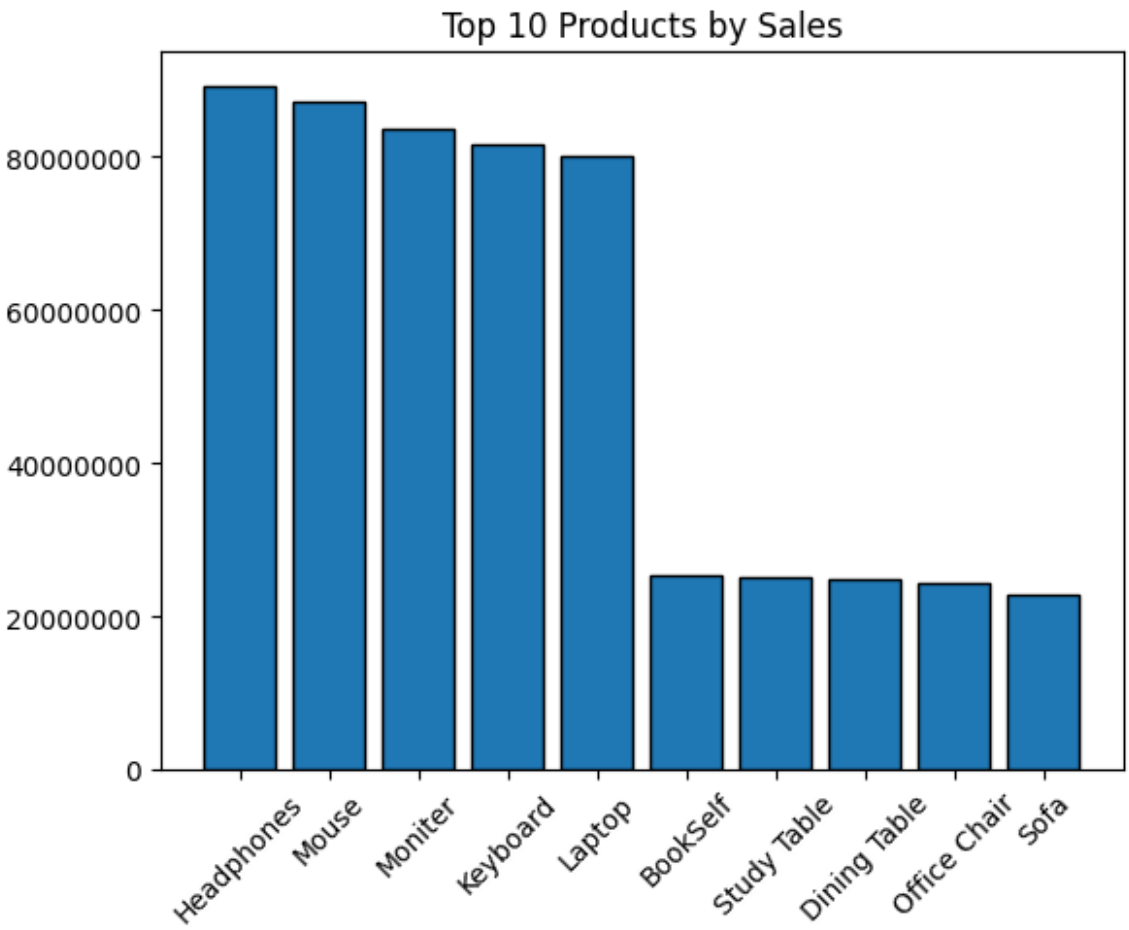
```
plt.bar_label(bar1)


delivery_counts = df["Delivery Status"].value_counts()

delivery_counts
plt.subplot(2,2,2)
plt.pie(delivery_counts.values, labels=delivery_counts.index,
autopct="%1.1f%%")
plt.title("Delivery Status Distribution")
plt.show()
```





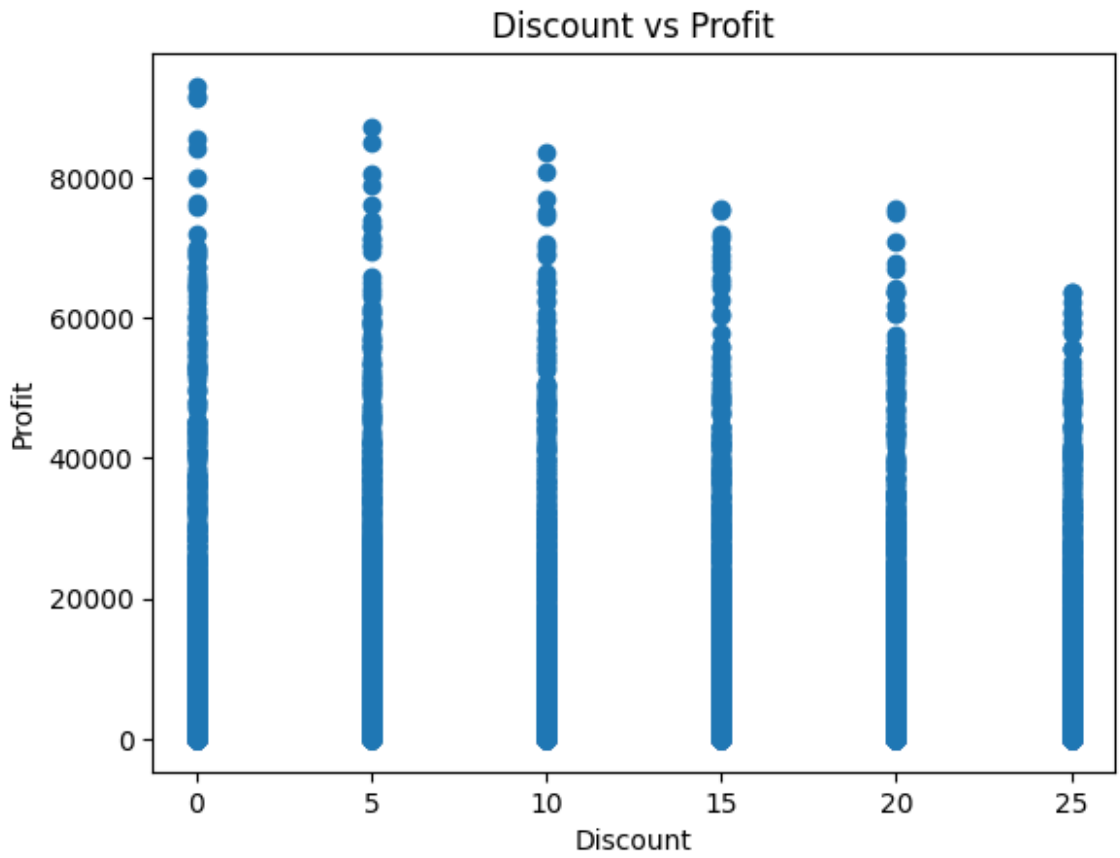
Top prouct Analysis

```
top_products = df.groupby("Product Name")["Sales
Amount"].sum().sort_values(ascending=False).head(10)


plt.figure()
plt.bar(top_products.index, top_products.values, edgecolor="black")
plt.title("Top 10 Products by Sales")
plt.xticks(rotation=45)
plt.ticklabel_format(style='plain', axis='y')
plt.show()
```



Discount Impact Analysis

```
plt.figure()
plt.scatter(df["Discount (%)"], df["Profit"])
plt.title("Discount vs Profit")
plt.xlabel("Discount")
plt.ylabel("Profit")
plt.show()
```



There is a negative relationship between discount and profit.

When discount increases, profit goes down.